Group Assignment ESS, part II

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Agenda

1. Introduction: selected country, sample period, data and LLM
2. Descriptive Analysis of the variables
3. Alternative Model Specifications and Hypotheses on Parameters
4. Diagnostic Checks on selected model
5. Forecast comparison: D4 vs AR(2) vs VAR(2)
6. Forecast comparison: D4 vs AR(2) vs VAR(2)

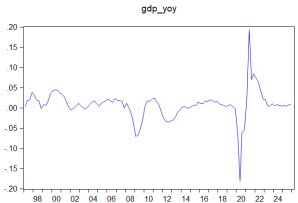
Selected country, sample period and data

- Selected country: **Italy**
- **Sample period**: **1999Q1–2019Q4**, we chose this period both to avoid the early 90's (troubles in the managed exchange system, structural and policy changes with the adoption of Maastricht Treaty) and Covid period. Hence, we start our analysis in 1999 and end it in 2019 at the cost of losing estimation efficiency due to rather short sample. Still, we plot and perform descriptive analysis of all variables in a larger sample (1997Q1–2026Q1).
- **Estimation sample**: **1999Q1–2009Q4**
- **Forecast evaluation**: **2010Q1–2019Q4** $\rightarrow$ 10 years, as required by point 5. of the assignment
- Following Ghysels and Marcellino (Chapter 1), alternative specifications can be compared in–sample (1999Q1–2019Q4) using significance, fit and information criteria. However, since the assignment requires (point 5.) an out–of–sample evaluation over the last 10 years of the sample, model selection (point 2.) is conducted on the estimation sample only.

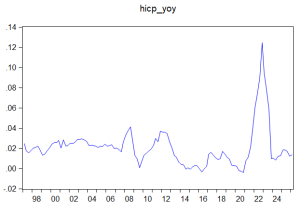
Data and LLM usage

- **Data** $\rightarrow$ Following Ghysels and Marcellino, we use real GDP (gdp), an harmonized measure of Italian inflation (hicp), industrial production (ip), a soft indicator for Italy (esi) and a financial variable reflecting the Italian stock market (ftse mib). Monthly variables are hicp, esi and euribor (short–term 3–months interest rate, used in Q.6), but we aggregate them to obtain quarterly variables. Thus, all variables are quarterly and we compute their growth rates. Growth rates are not neutral, as they generally allow to mitigate serial correlation problems. All data are from Eurostat and DG ECFIN. Notice also that results in this deck are from data not multiplied by 100, while results in the Eviews workfile are multiplied by 100 as they were made by different people in the group: this does not induce any change in our conclusions, but only changes the magnitude of some numbers.
- **LLM usage** $\rightarrow$ We used an LLM (ChatGPT) which helped us in performing the following tasks: i) aggregation of monthly variables on Excel (carefully checked); ii) writing the code after the menu–driven analysis (carefully checked); iii) final check of any error in the slide deck.

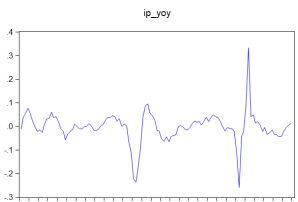
Time-series: GDP, inflation and industrial production



Real GDP growth

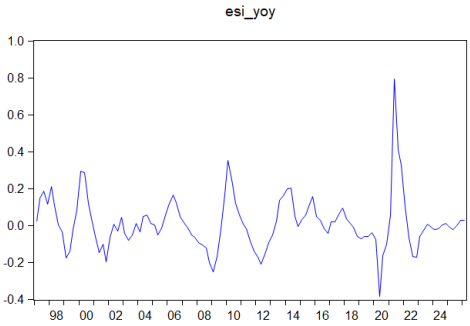


HICP inflation

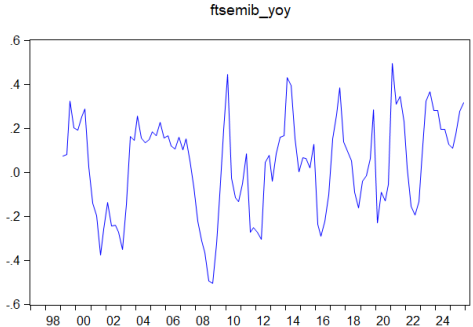


Industrial production

Time-series: survey and financial indicators



ESI Italy



FTSE MIB

Summary Statistics I

Table: Summary statistics

	GDP	HICP	IP	ESI	FTSE MIB
Mean	0.710178	2.087664	-0.374246	1.281098	3.389088
Std. Dev.	3.507082	1.937929	6.384778	14.53332	21.91750
Min	-18.13402	-0.399202	-25.94433	-38.54712	-50.64535
Max	19.31833	12.43243	33.28859	79.55272	49.35048
Skewness	-0.222550	2.543296	-0.208938	1.514986	-0.292952
Kurtosis	15.54186	12.01236	12.65136	9.636316	2.414696
Obs.	117	117	117	117	109

Notes: Sample: 1997Q1–2026Q1. Quarterly year-on-year growth rates. FTSEMIB only defined from 1999Q1.

Summary Statistics II

Table: Correlation matrix

	GDP	HICP	IP	ESI	FTSE MIB
GDP	1.000000	0.245244	0.855675	0.682679	0.330636
HICP	0.245244	1.000000	0.033101	-0.175248	-0.132161
IP	0.855675	0.033101	1.000000	0.710253	0.395200
ESI	0.682679	-0.175248	0.710253	1.000000	0.558875
FTSEMIB	0.330636	-0.132161	0.395200	0.558875	1.000000

Notes: Sample: 1997Q1–2026Q1. Quarterly year-on-year growth rates. FTSEMIB only defined from 1999Q1.

Model Specifications

Following Ghysels and Marcellino (Chapter 1), we consider a limited set of alternative forecasting models for GDP growth, combining *hard*, *soft* and *financial* indicators.

We first estimate **static specifications** with different subsets of regressors. We then introduce **dynamic ARDL specifications** to account for persistence in GDP growth and for possible lagged effects of the predictors: we use just one lag as in Ghysels and Marcellino (Chapter 3).

The preferred specification, D4, is selected according to coefficient significance, model fit, information criteria, parsimony and residual diagnostics.

- $M1 \rightarrow$ Static general: IP + soft + financial + inflation
- $M2 \rightarrow$ Static reduced: IP + soft + financial
- $M3 \rightarrow$ Static parsimonious: IP + soft
- $D1 \rightarrow$ ARDL general
- $D2 \rightarrow$ ARDL reduced
- $D3 \rightarrow$ ARDL parsimonious
- $D4 \rightarrow$ **Final model!**

Static model specifications: M1-M2-M3

	M1	M2	M3
Constant	0.0187***	0.0123***	0.0123***
	(0.0065)	(0.0013)	(0.0013)
IP_YOY	0.3698***	0.3448***	0.3492***
	(0.0357)	(0.0256)	(0.0242)
ESI_YOY	0.0123	0.0126	0.0171
	(0.0153)	(0.0153)	(0.0131)
FTSEMIB_YOY	-0.0011	0.0046	
	(0.0096)	(0.0078)	
HICP_YOY	-0.2692		
	(0.2680)		
Adj. R^2	0.8821	0.8820	0.8839
AIC	-6.5661	-6.5860	-6.6229
SIC	-6.3633	-6.4237	-6.5012
HQ	-6.4908	-6.5258	-6.5777
Obs.	44	44	44

Notes: Standard errors in parentheses. ***

$p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Dynamic model specifications: D1-D2-D3-D4

	D1	D2	D3	D4
Constant	0.0054	0.0172***	0.0080***	0.0024
	(0.0057)	(0.0048)	(0.0016)	(0.0015)
GDP_YOY(-1)	0.7722***	0.4414***	0.3565***	0.7979***
	(0.1047)	(0.0863)	(0.0894)	(0.1003)
IP_YOY	0.2396***	0.2144***	0.2239***	0.3120***
	(0.0444)	(0.0401)	(0.0362)	(0.0286)
IP_YOY(-1)	-0.1919***			-0.2410***
	(0.0598)			(0.0422)
ESI_YOY	0.0007			
	(0.0161)			
ESI_YOY(-1)	0.0095	0.0091	0.0198	
	(0.0152)	(0.0128)	(0.0133)	
FTSEMIB_YOY	0.0025			
	(0.0084)			
FTSEMIB_YOY(-1)	0.0066	0.0063		
	(0.0093)	(0.0080)		
HICP_YOY	0.1428	-0.4450**		
	(0.2426)	(0.2026)		
HICP_YOY(-1)	-0.2730			
	(0.2326)			

	D1	D2	D3	D4
Adj. R^2	0.9548	0.9327	0.9175	0.9521
AIC	-7.4073	-7.0813	-6.9440	-7.4865
SIC	-6.9977	-6.8355	-6.7818	-7.3243
HQ	-7.2563	-6.9907	-6.8838	-7.4264
Obs.	43	43	44	44

Notes: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$; values for FTSEMIB-YOY only defined from 1999q1, thus D1 and D2 are estimated on a sample starting in 1999q2.

Discussion: Static Models

- Among the static specifications, M3 is the preferred model. Industrial production is the only robust and highly significant predictor across all models, while financial and inflation variables do not add explanatory power. M3 also dominates in terms of adjusted R^2 and information criteria, making it the most parsimonious and best-performing static specification.
- Notice also that in M1 negative signs are found on the estimated parameters for the change of FTSE MIB and the inflation rate, contrary to economic intuition. This result can be due either to collinearity among the regressors or to model misspecification. We therefore exclude first inflation (M2), and then FTSE MIB (M3).
- In conclusion, our results are consistent also over a shorter sample estimation period (1999Q1–2006Q4, 32 observations) that excludes the 2008 crisis, even though in this estimation exercise the values of adjusted R^2 are lower (as in Ghysels and Marcellino, Chapter 1).

Discussion: Dynamic Models and Hypotheses on parameters

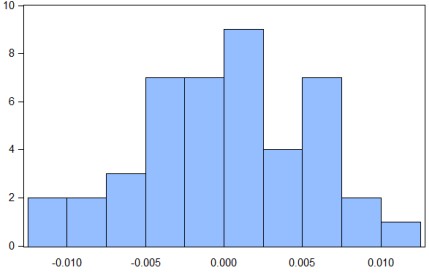
- All three dynamic specifications confirm that GDP growth is strongly persistent: the coefficient on `GDP_YOY(-1)` is positive and highly significant in every model.
- Industrial production is the most robust explanatory variable across specifications. `IP_YOY` is always positive and highly significant, while in D1 the lagged term `IP_YOY(-1)` is also significant and negative, suggesting a richer short-run dynamic structure.
- The soft indicator, the financial variable and inflation do not appear to add robust explanatory power. Particularly, inflation provides mixed evidence. `HICP_YOY` is significant only in D2 and with a negative coefficient, but it is not significant in the more general specification D1; therefore, its role should be interpreted with caution.
- In terms of overall fit, D1 clearly dominates: it has the highest adjusted R^2, the lowest AIC, SIC and HQ, and the best Durbin–Watson statistic. This suggests that D1 is the preferred dynamic specification among the three models.
- At the same time, D1 also shows that several additional regressors are not statistically significant (overfitting). D1 has the best in-sample fit, but D4 achieves a better trade-off between parsimony and information criteria: thus, **we adopt D4**.

Diagnostic Checks I

- Both the **White test** and the **Breusch–Pagan–Godfrey** test yield large p-values, so the null of homoskedasticity cannot be rejected for model D4 at conventional significance levels. **Breusch–Godfrey test** for serial correlation does not reject the null of no serial correlation up to 4 lags (we use quarterly data, so as a rule of thumb 4 lags should be natural). The **Jarque–Bera test** does not reject residual normality.
- We proceed by first using the **Chow breakpoint test** in the full sample (1999Q1–2019Q4): in principle, we should first test that the error variance does not change over the two subperiods, but since we did not reject homoskedasticity we take the variance to be constant. Following Ghysels and Marcellino (Chapter 2), we test 2008Q4 and 2009Q1 as single breakpoints: we find no evidence of parameter instability.
- We also use **recursive methods** to look into the recursive residuals: overall, the evidence does not support a sharp breakpoint, but it suggests some mild gradual parameter instability over time.

Diagnostic Checks II

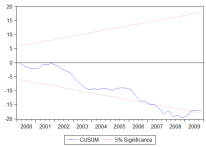
Test	Sample	*p*-value
BPG homoskedasticity	1999Q1–2009Q4	0.7872
White homoskedasticity	1999Q1–2009Q4	0.9663
BG LM(4) serial corr.	1999Q1–2009Q4	0.3367
Chow at 2008Q4	1999Q1–2019Q4	0.3755
Chow at 2009Q1	1999Q1–2019Q4	0.1997

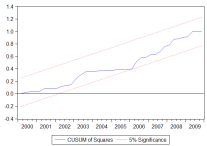


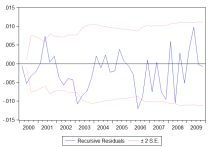
Series: Residuals
Sample 1999Q1 2009Q4
Observations 44

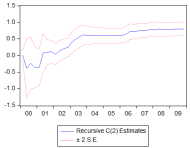
Mean	-7.27e-19
Median	7.27e-05
Maximum	0.011328
Minimum	-0.012354
Std. Dev.	0.005292
Skewness	-0.148390
Kurtosis	2.630458
Jarque-Bera	0.411838
Probability	0.813899

Diagnostic Checks III

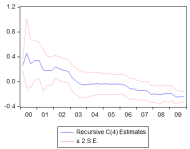




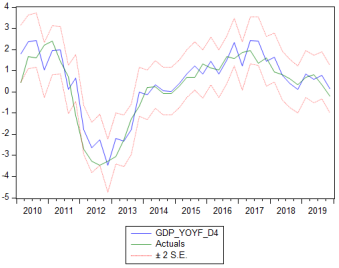




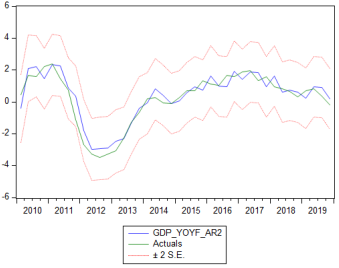




Forecast comparison: D4 vs AR(2) vs VAR(2)

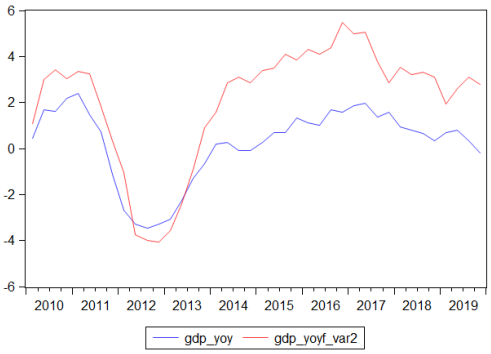


Forecast: GDP_YOYF_D4	
Actual: GDP_YOY	
Forecast sample: 2010Q1 2019Q4	
Included observations: 40	
Root Mean Squared Error	0.643893
Mean Absolute Error	0.509110
Mean Abs. Percent Error	61.87589
Theil Inequality Coef.	0.202658
Bias Proportion	0.128715
Variance Proportion	0.016799
Covariance Proportion	0.854486
Theil U2 Coefficient	0.662093
Symmetric MAPE	64.19367



Forecast: GDP_YOYF_AR2	
Actual: GDP_YOY	
Forecast sample: 2010Q1 2019Q4	
Included observations: 40	
Root Mean Squared Error	0.511406
Mean Absolute Error	0.415279
Mean Abs. Percent Error	66.25868
Theil Inequality Coef.	0.164105
Bias Proportion	0.074128
Variance Proportion	0.055957
Covariance Proportion	0.869915
Theil U2 Coefficient	0.785583
Symmetric MAPE	55.66256

Forecast comparison: D4 vs AR(2) vs VAR(2)



Model	**RMSFE**	**MAFE**
AR(2)	0.5114	0.4153
D4	0.6439	0.5091
VAR(2)	2.2400	1.9962

Notes: Forecast evaluation refers to GDP_YOY over 2010Q1–2019Q4.

- Following Ghysels and Marcellino, forecast accuracy is evaluated primarily through RMSFE and MAFE. The relatively poor forecast performance of VAR(2) is not surprising. The VAR is more heavily parameterized, is estimated on a short sample, and does not include industrial production, which is the most informative predictor of GDP growth in our analysis.